

1. Questions

There are two series given below. Series I has one wrong term and series II starts with wrong term of series I. Both series follows same pattern.

Series I: 10, 18, 45, 70, 413, 534

Series II: _____, __[A]____, _____, _____, _____, __[B]____

Find the value of $\{[\sqrt{(B-1)}]+A\}$.

- a. 45
- b. 37
- c. 29
- d. 18
- e. None of these

2. Questions

There three series given below. Series I, II and III has one missing term as P, Q and R respectively. Each series follows different pattern.

Series I: 20, 42, 72, [P], 156, 210

Series II: 11, [Q], 39, 74, 123, 186

Series III: 32, 26, 38, 14, 18, [R]

Find the correct relation between P, Q and R.

- a. $P > Q > R$
- b. $P < Q < R$
- c. $P > Q < R$
- d. $P < Q > R$
- e. $P = Q = R$

3. Questions

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Read the following information carefully and answer the questions based on it.

The table chart given below the total number of questions attempted (correct + wrong) by a student in four different subjects A, B, C and D. It also shows the percentage of the number of correct questions attempted out of the total number of questions attempted and the total number of marks obtained by the same student in each subject.

Subject	Total number of questions attempted (correct + wrong)	Percentage of the number of correct questions attempted	Total number of marks obtained
A	20P	85%	(16P+0.75)
B	72	15Q%	(10Q-0.50)
C	20R	20R%	60
D	45S	70%	56.25

Note:

The values of each variable is positive integer.

Each subject consists of 100 questions.

In subject B, the number of correct questions attempted is (7Q+1) more than that of wrong questions attempted.

In subject C number of wrong questions attempted is 48 less than that of correct questions attempted.

A student will be awarded 1 mark for each correct attempt and 0.25 marks will be deducted for each wrong attempt.

No deduction for number of skipped questions.

Total number of marks obtained = (number of marks scored from correct attempts) – (number of marks deducted for wrong attempts)

Total number of questions = number of questions attempted + number of questions skipped

Total number of attempted questions = number of correct attempts + number of wrong attempts

In subject D, there are two sections theory and practical. If the ratio of the number of correct questions attempted in theory section to that of wrong questions attempted in practical section is 25:1. And, the number of correct questions attempted in practical section is 48% less than that of wrong questions attempted in practical section, then find the number of wrong questions attempted in practical section.

- 2
- 5
- 10
- 12
- None of these

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4. Questions

In subject A, if 2 marks is awarded for each correct attempt and 0.50 mark is deducted for each wrong attempt, then find the percentage change in the total number of marks obtained as compared to the original number of total marks obtained by the student in subject A.

- a. 100% decreased
- b. 100% increased
- c. 50% increased
- d. 50% decreased
- e. None of these

5. Questions

If there is another subject E, in which a student has obtained the same number of total marks as in subject D and the total number of questions attempted by him subject E is 75, then find the difference between the number of correct and wrong attempts made by the student in subject E.

- a. 40
- b. 35
- c. 20
- d. Cannot be determined
- e. None of these

6. Questions

If there is another student, whose number correct attempts in subject B is 4 less than that of the present student and his number of wrong attempts is 2 more than that of the present student in the same subject, then find the total number of marks obtained by another student in subject B.

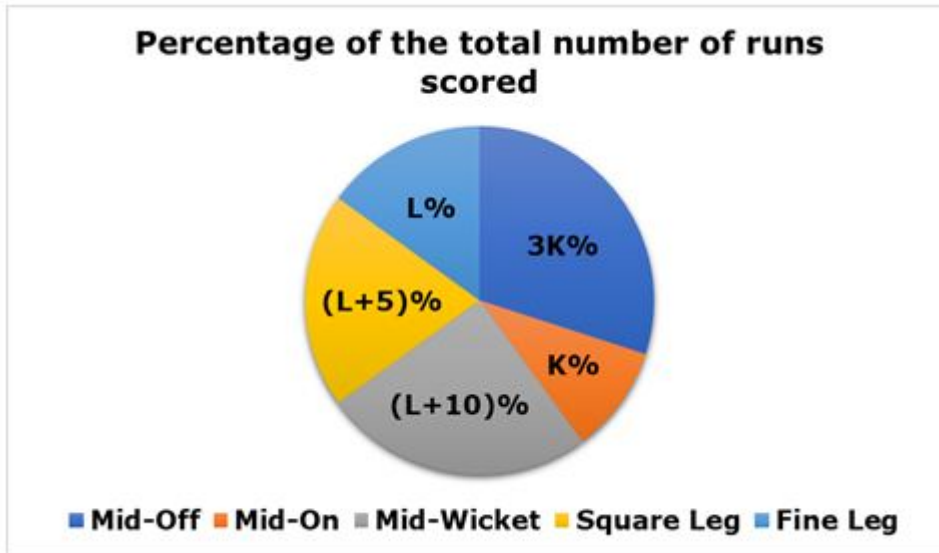
- a. 32
- b. 45
- c. 26
- d. 50
- e. None of these

7. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored by a player among 11 players of a particular team in five different fielding positions i.e., mid-off, mid-on, mid-wicket, square-leg and fine leg during a cricket match.

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Note:

The ratio of the number of runs scored by the player in mid-off to that of square-leg is 3:2.

The number of runs scored in mid-wicket is (K+L+10).

A team consists a total of 11 players.

Strike rate = $\left[\frac{\text{(number of runs scored)}}{\text{(number of balls faced)}} \times 100 \right]$

The number of runs scored can scored as singles (1 run), doubles (2 runs), triple (3 runs), fours (4 runs) and sixes (6 runs).

The number of runs scored in boundaries = number of fours scored + number of sixes scored

The dot ball faced by a player means zero (0) number of runs scored on that particular ball by that player.

Total number of runs scored by a team of 11 players = number of runs scored by that particular player + number of runs scored by the remaining 10 players

If the total number of scored by a player is X% of the total number of runs scored by all the 11 players together (including that player also). Then find the value of 2X, if the average number of runs scored by the remaining 10 players is 26.

- a. 35
- b. 70
- c. 21
- d. 42
- e. None of these

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8. Questions

If the total number of balls faced by the player is B and his overall strike rate is (B+130), then find the difference between the total number runs scored and the overall strike rate of the same player. (B should be positive integer)

- a. 80
- b. 100
- c. 60
- d. Cannot be determined
- e. None of these

9. Questions

Out of the total number of runs scored by the player in mid-off, the number of singles, doubles and triples scored are 13, 1 and 1 respectively. If the total number of boundaries (fours + sixes) scored is 5, then find the total number of runs scored in mid-off through sixes only.

- a. 12
- b. 18
- c. 16
- d. 14
- e. None of these

10. Questions

A player has a strike rate of 210 in fine leg and the number of dot balls faced by the same player in the fine leg position is 3. Find his strike rate in the fine leg position, if the number of dot balls faced by him is excluded.

- a. 320
- b. 280
- c. 240
- d. 300
- e. None of these

11. Questions

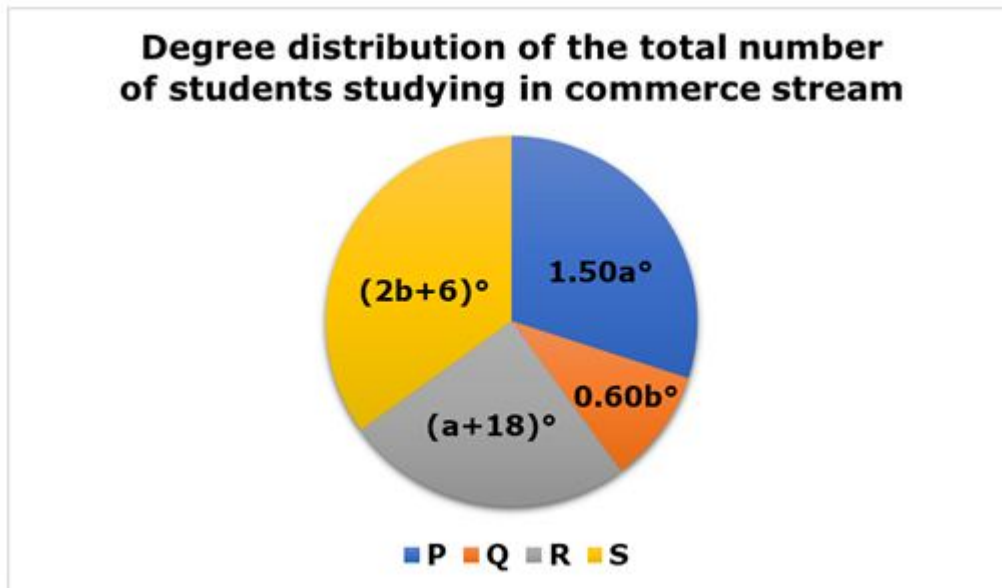
The ratio of the total number of balls faced to that of the overall strike rate of another player of the same team is 5:16 and the total number of runs scored by that player is 42.85% less than that of present player, then find the total number of balls faced by another player.

- a. 50
- b. 60
- c. 40
- d. 30
- e. None of these

12. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of students studying in commerce stream in four different colleges P, Q, R and S.



Note:

The difference between the values of a and b is positive root of the equation $P^2 - 3P - 108 = 0$. ($a > b$)

The average number of students studying in commerce stream in college P and college Q is 40.

Total number of students studying = number of students studying science + number of students studying commerce + number of students studying on arts

Total number of students studying = number of boys studying + number of girls studying

In college P, the total number of students studying in commerce stream is X% of that of arts stream and the number of students studying in science stream is (X-48), which is 62 more than that of arts stream, then find the total number of students studying in all three streams together in college P. (X is a positive integer)

- a. 184
- b. 158
- c. 202
- d. Cannot be determined
- e. None of these

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13. Questions

In college S in commerce stream, the number of boys studying is 8Y, which is Z more than that of girls studying in the same stream of same college. If the number of girls studying in commerce stream in college S is 3Z, then find the value of (Y:Z).

- a. 2:1
- b. 1:2
- c. 3:2
- d. 2:3
- e. None of these

14. Questions

The total number of students studying in all the three stream (science + commerce + arts) together in college Q is 81 and the number of students studying science stream is 1 more than five times the difference between the number of students studying in commerce and arts stream, then find the number of students studying in science stream in college Q. (number of commerce students studying is more than that of arts students studying in college Q)

- a. 42
- b. 37
- c. 23
- d. 51
- e. None of these

15. Questions

If there is another college T, where the number of students studying in commerce stream is K, which is $\frac{1}{6}^{\text{th}}$ of the total number of students studying in all the five colleges together (including college T), then find the number of students studying in commerce stream in college T.

- a. 40
- b. 33
- c. 27
- d. 15
- e. None of these

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16. Questions

Following question have two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give an answer accordingly.

A and B together can complete a piece of work in $(40/7)$ days, whereas C and D together can complete the same piece of work in $(60/11)$ days. Efficiency of A is 40% less than that of D and the ratio of the efficiency of B to that of C is 5:4.

Quantity I: The time taken by A and C to complete the work together.

Quantity II: The time taken by B and D to complete the work together.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

17. Questions

The downstream and upstream speed of the boat are $(7X+2)$ km/h and (X^2+2) km/h respectively. The ratio of the time taken by the boat to cover 240 km in downstream to that of 180 km in upstream is 4:5. (X is a positive integer)

Quantity I: The speed of the boat in still water is what percent of that of the speed of the river?

Quantity II: The downstream distance covered by the boat in 7 hours.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

18. Questions

Cost price of the shirt is Rs. 500, which is sold at profit of 20%. Marked price of the jeans is Rs. 370 more than that of shirt and jeans is marked-up 40% above its cost price. Selling price of the shirt is $\frac{3}{4}$ th of the cost price of the jeans and jeans is sold at a discount of 10%.

Quantity I: Marked price of shirt.

Quantity II: Selling price of jeans.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

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19. Questions

A person invested Rs. 40000 at the rate of 5R% per annum simple interest for R years and received Rs. 18000 as interest.

Quantity I: Find the compound interest received after 2 years, if he had invested Rs. 30000 at the $(R+7)\%$ per annum compounded annually.

Quantity II: If Rs. 8000R is invested at the rate of 18% per annum simple interest, then interest received after 4 years.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

20. Questions

The ratio of the income of A to that of B is 7:6 and the expenditure of B is 10% less than that of A.

The savings of A is $\frac{1}{3}^{\text{rd}}$ more than that of B. The income of A is Rs. 10000 more than the expenditure of B and the income of B is Rs. 4000 more than the expenditure of A.

Quantity I: The difference between the expenditure and savings of B.

Quantity II: The total income (expenditure + savings) of A.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

Explanations:

1. Questions

Answer: B

Series I: 10, 18, 45, 70, 413, 534

6th term = 534

5th term = $(534 - 11^2) = (534 - 121) = 413$

4th term = $(413 - 7^3) = (413 - 343) = 70$

3rd term = $(70 - 5^2) = (70 - 25) = 45$

2nd term = $(45 - 3^3) = (45 - 27) = 18$

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$$1^{\text{st}} \text{ term} = (18-2^2) = (18-4) = 14 \text{ (10 is wrong)}$$

So, the wrong term = 10

And, the correct term = 14

Series II:

$$1^{\text{st}} \text{ term} = 10$$

$$2^{\text{nd}} \text{ term} = (10+2^2) = (10+4) = 14$$

$$3^{\text{rd}} \text{ term} = (14+3^3) = (14+27) = 41$$

$$4^{\text{th}} \text{ term} = (41+5^2) = (41+25) = 66$$

$$5^{\text{th}} \text{ term} = (66+7^3) = (66+343) = 409$$

$$6^{\text{th}} \text{ term} = (409+11^2) = (409+121) = 530$$

So, A = 14

And, B = 530

Hence, the value of,

$$[\{\sqrt{(B-1)}\}+A] = [\{\sqrt{(530-1)}\}+14] = [\{\sqrt{(529)}\}+14] = (23+14) = 37.$$

2. Questions

Answer: A

Series I: 20, 42, 72, [P], 156, 210

$$1^{\text{st}} \text{ term} = (4 \times 5) = 20$$

$$2^{\text{nd}} \text{ term} = (6 \times 7) = 42$$

$$3^{\text{rd}} \text{ term} = (8 \times 9) = 72$$

$$4^{\text{th}} \text{ term} = (10 \times 11) = 110$$

$$5^{\text{th}} \text{ term} = (12 \times 13) = 156$$

$$6^{\text{th}} \text{ term} = (14 \times 15) = 210$$

So, P = 110

Series II: 11, [Q], 39, 74, 123, 186

$$6^{\text{th}} \text{ term} = 186$$

$$5^{\text{th}} \text{ term} = \{186-(7 \times 9)\} = (186-63) = 123$$

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$$4^{\text{th}} \text{ term} = \{123 - (7 \times 7)\} = (123 - 49) = 74$$

$$3^{\text{rd}} \text{ term} = \{74 - (7 \times 5)\} = (74 - 35) = 39$$

$$2^{\text{nd}} \text{ term} = \{39 - (7 \times 3)\} = (39 - 21) = 18$$

$$1^{\text{st}} \text{ term} = \{18 - (7 \times 1)\} = (18 - 7) = 11$$

$$\text{So, } Q = 18$$

Series III: 32, 26, 38, 14, 18, [R]

$$1^{\text{st}} \text{ term} = 32$$

$$2^{\text{nd}} \text{ term} = \{32 - (3 \times 2)\} = (32 - 6) = 26$$

$$3^{\text{rd}} \text{ term} = \{26 + (2 \times 6)\} = (26 + 12) = 38$$

$$4^{\text{th}} \text{ term} = \{38 - (3 \times 8)\} = (38 - 24) = 14$$

$$5^{\text{th}} \text{ term} = \{14 + (1 \times 4)\} = (14 + 4) = 18$$

$$6^{\text{th}} \text{ term} = \{18 - (1 \times 8)\} = (18 - 8) = 10$$

$$\text{So, } R = 10$$

Now,

$$P = 110$$

$$Q = 18$$

$$R = 10$$

Hence, $P > Q > R$.

3. Questions

Subject A

The total number of questions attempted = $20P$

The number of correct questions attempted = 85% of $20P = 17P$ @QuizByMickey

The number of wrong questions attempted = 15% of $20P = 3P$

So, the total number of marks scored from correct attempts = $(17P \times 1) = 17P$

And, the total number of marks deducted from wrong attempts = $(3P \times 0.25) = 0.75P$

Therefore, the total number of marks obtained = $(17P - 0.75P) = 16.25P$

Now,

$$16.25P = (16P + 0.75)$$

$$0.25P = 0.75$$

$$P = 3$$

So, the total number of questions attempted = $20P = 60$

The number of correct questions attempted = $17P = 51$

The number of wrong questions attempted = $3P = 9$

Therefore, the total number of marks scored from correct attempts = $17P = 51$

And, the total number of marks deducted from wrong attempts = $0.75P = 2.25$

Therefore, the total number of marks obtained = $16.25P = 48.75$

Subject B

The total number of questions attempted = 72

The number of correct attempts = $15Q\%$ of 72 = $10.80Q$

So, the number of wrong attempts = $(72 - 10.80Q)$

Now,

$$10.80Q - (72 - 10.80Q) = (7Q + 1)$$

$$3.80Q - 1 = 72 - 10.80Q$$

$$14.60Q = 73$$

$$Q = 5$$

Therefore, the total number of questions attempted = 72

The number of correct attempts = $10.80Q = 54$

And, the number of wrong attempts = $(72 - 10.80Q) = (72 - 54) = 18$

The total number of marks scored from correct attempts = $(54 \times 1) = 54$

And, the total number of marks deducted for wrong attempts = $(18 \times 0.25) = 4.50$

So, the total marks obtained = $(54 - 4.50) = 49.50$

Subject C

The total number of questions attempted = $20R$

The number of correct attempts = $20R\%$ of $20R = 4R^2$

So, the number of wrong attempts = $(20R - 4R^2)$

Now,

$$4R^2 - (20R - 4R^2) = 48$$

$$8R^2 - 20R - 48 = 0$$

$$2R^2 - 5R - 12 = 0$$

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$$2R^2 - 8R + 3R - 12 = 0$$

$$2R(R-4) + 3(R-4) = 0$$

$$(2R+3)(R-4) = 0$$

$$R = -1.50 \text{ or } 4$$

So, $R = 4$ (positive integer)

Therefore, the total number of questions attempted $= 20R = 80$

The number of correct attempts $= 20R\%$ of $20R = 4R^2 = 64$

And, the number of wrong attempts $= (20R - 4R^2) = (80 - 64) = 16$

Now,

The total number of marks scored from correct attempts $= (64 \times 1) = 64$

And, the total number of marks deducted for wrong attempts $= (16 \times 0.25) = 4$

So, the total marks obtained $= (64 - 4) = 60$

Subject D

The total number of questions attempted $= 45S$

The number of correct attempts $= 70\%$ of $45S = 31.50S$

So, the number of wrong attempts $= (45S - 31.50S) = 13.50S$

Now,

The total number of marks scored from correct attempts $= (31.50S \times 1) = 31.50S$

And, the total number of marks deducted for wrong attempts $= (13.50S \times 0.25) = 3.375S$

So, the total marks obtained $= (31.50S - 3.375S) = 28.125S$

$$28.125S = 56.25$$

$$S = 2$$

Therefore, the total number of questions attempted $= 45S = 90$

The number of correct attempts $= 31.50S = 63$

So, the number of wrong attempts $= 13.50S = 27$

And, the total number of marks scored from correct attempts $= 31.50S = 63$

And, the total number of marks deducted for wrong attempts $= 3.375S = 6.75$

So, the total marks obtained $= (63 - 6.75) = 56.25$

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Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: A

According to the questions,

Subject D

Let, the total number of correct questions attempted in theory section and wrong questions attempted in practical section be $25m$ and m respectively.

And, the ratio of the number of correct questions attempted in practical section to that of wrong questions attempted in theory section = $52:100 = 13:25$. Let, it be $13n$ and $25n$ respectively.

The total number of correct questions attempted = 63

The total number of wrong questions attempted = 27

Now,

$$25m + 13n = 63 \dots\dots\dots(i)$$

$$m + 25n = 27 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$25m + 13n = 63$$

$$25m + 625n = 675$$

$$612n = 612$$

$$n = 1$$

Now, put the value of n in equation (i), we get,

$$25m + 13n = 63$$

$$25m + 13 = 63$$

$$25m = 50$$

$$m = 2$$

Hence, the number of wrong questions attempted in practical section = $m = 2$.

4. Questions

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Subject A

The total number of questions attempted = 20P

The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = $(17P \times 1) = 17P$

And, the total number of marks deducted from wrong attempts = $(3P \times 0.25) = 0.75P$

Therefore, the total number of marks obtained = $(17P - 0.75P) = 16.25P$

Now,

$$16.25P = (16P + 0.75)$$

$$0.25P = 0.75$$

$$P = 3$$

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = $(72 - 10.80Q)$

Now,

$$10.80Q - (72 - 10.80Q) = (7Q + 1)$$

$$3.80Q - 1 = 72 - 10.80Q$$

$$14.60Q = 73$$

$$Q = 5$$

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = $(72 - 10.80Q) = (72 - 54) = 18$

The total number of marks scored from correct attempts = $(54 \times 1) = 54$

And, the total number of marks deducted for wrong attempts = $(18 \times 0.25) = 4.50$

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So, the total marks obtained = $(54 - 4.50) = 49.50$

Subject C

The total number of questions attempted = $20R$

The number of correct attempts = $20R\%$ of $20R = 4R^2$

So, the number of wrong attempts = $(20R - 4R^2)$

Now,

$$4R^2 - (20R - 4R^2) = 48$$

$$8R^2 - 20R - 48 = 0$$

$$2R^2 - 5R - 12 = 0$$

$$2R^2 - 8R + 3R - 12 = 0$$

$$2R(R-4) + 3(R-4) = 0$$

$$(2R+3)(R-4) = 0$$

$$R = -1.50 \text{ or } 4$$

So, $R = 4$ (positive integer)

Therefore, the total number of questions attempted = $20R = 80$

The number of correct attempts = $20R\%$ of $20R = 4R^2 = 64$

And, the number of wrong attempts = $(20R - 4R^2) = (80 - 64) = 16$

Now,

The total number of marks scored from correct attempts = $(64 \times 1) = 64$

And, the total number of marks deducted for wrong attempts = $(16 \times 0.25) = 4$

So, the total marks obtained = $(64 - 4) = 60$

Subject D

The total number of questions attempted = $45S$

The number of correct attempts = 70% of $45S = 31.50S$

So, the number of wrong attempts = $(45S - 31.50S) = 13.50S$

Now,

The total number of marks scored from correct attempts = $(31.50S \times 1) = 31.50S$

And, the total number of marks deducted for wrong attempts = $(13.50S \times 0.25) = 3.375S$

So, the total marks obtained = $(31.50S - 3.375S) = 28.125S$

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$$28.125S = 56.25$$

$$S = 2$$

Therefore, the total number of questions attempted = $45S = 90$

The number of correct attempts = $31.50S = 63$

So, the number of wrong attempts = $13.50S = 27$

And, the total number of marks scored from correct attempts = $31.50S = 63$

And, the total number of marks deducted for wrong attempts = $3.375S = 6.75$

So, the total marks obtained = $(63 - 6.75) = 56.25$

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
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D	90	63	27	56.25
Total	302	232	70	214.50

Answer: B

Subject A

The total number of correct attempts = 51

The total number of wrong attempts = 9

So, the number of marks obtained from correct attempts = $(51 \times 2) = 102$

And, the number of marks deducted from wrong attempts = $(9 \times 0.50) = 4.50$

Therefore, the new number of total marks obtained = $(102 - 4.50) = 97.50$

And, the original number of total marks obtained = 48.75

Hence, the required % = $[\{(97.50 - 48.75) / 48.75\} \times 100]\% = \{(48.75 / 48.75) \times 100\}\% = 100\%$ increased.

5. Questions

Subject A

The total number of questions attempted = $20P$

The number of correct questions attempted = 85% of $20P = 17P$

The number of wrong questions attempted = 15% of $20P = 3P$

So, the total number of marks scored from correct attempts = $(17P \times 1) = 17P$

And, the total number of marks deducted from wrong attempts = $(3P \times 0.25) = 0.75P$

Therefore, the total number of marks obtained = $(17P - 0.75P) = 16.25P$

Now,

$$16.25P = (16P + 0.75)$$

$$0.25P = 0.75$$

$$P = 3$$

So, the total number of questions attempted = $20P = 60$

The number of correct questions attempted = $17P = 51$

The number of wrong questions attempted = $3P = 9$

Therefore, the total number of marks scored from correct attempts = $17P = 51$

And, the total number of marks deducted from wrong attempts = $0.75P = 2.25$

Therefore, the total number of marks obtained = $16.25P = 48.75$

Subject B

The total number of questions attempted = 72

The number of correct attempts = $15Q\%$ of 72 = $10.80Q$

So, the number of wrong attempts = $(72 - 10.80Q)$

Now,

$$10.80Q - (72 - 10.80Q) = (7Q + 1)$$

$$3.80Q - 1 = 72 - 10.80Q$$

$$14.60Q = 73$$

$$Q = 5$$

Therefore, the total number of questions attempted = 72

The number of correct attempts = $10.80Q = 54$

And, the number of wrong attempts = $(72 - 10.80Q) = (72 - 54) = 18$

The total number of marks scored from correct attempts = $(54 \times 1) = 54$

And, the total number of marks deducted for wrong attempts = $(18 \times 0.25) = 4.50$

So, the total marks obtained = $(54 - 4.50) = 49.50$

Subject C

The total number of questions attempted = $20R$

The number of correct attempts = $20R\%$ of $20R = 4R^2$

So, the number of wrong attempts = $(20R - 4R^2)$

Now,

$$4R^2 - (20R - 4R^2) = 48$$

$$8R^2 - 20R - 48 = 0$$

$$2R^2 - 5R - 12 = 0$$

$$2R^2 - 8R + 3R - 12 = 0$$

$$2R(R-4) + 3(R-4) = 0$$

$$(2R+3)(R-4) = 0$$

$$R = -1.50 \text{ or } 4$$

So, $R = 4$ (positive integer)

Therefore, the total number of questions attempted = $20R = 80$

The number of correct attempts = $20R\%$ of $20R = 4R^2 = 64$

And, the number of wrong attempts = $(20R - 4R^2) = (80 - 64) = 16$

Now,

The total number of marks scored from correct attempts = $(64 \times 1) = 64$

And, the total number of marks deducted for wrong attempts = $(16 \times 0.25) = 4$

So, the total marks obtained = $(64 - 4) = 60$

Subject D

The total number of questions attempted = $45S$

The number of correct attempts = 70% of $45S = 31.50S$

So, the number of wrong attempts = $(45S - 31.50S) = 13.50S$

Now,

The total number of marks scored from correct attempts = $(31.50S \times 1) = 31.50S$

And, the total number of marks deducted for wrong attempts = $(13.50S \times 0.25) = 3.375S$

So, the total marks obtained = $(31.50S - 3.375S) = 28.125S$

$$28.125S = 56.25$$

$$S = 2$$

Therefore, the total number of questions attempted = $45S = 90$

The number of correct attempts = $31.50S = 63$

So, the number of wrong attempts = $13.50S = 27$

And, the total number of marks scored from correct attempts = $31.50S = 63$

And, the total number of marks deducted for wrong attempts = $3.375S = 6.75$

So, the total marks obtained = $(63 - 6.75) = 56.25$

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: E

Subject E

The total number of questions attempted = 75

Let, the number of correct and wrong questions attempted be K and L respectively.

So, the number of marks obtained from correct attempts = $(K \times 1) = K$

And, the number of marks deducted for wrong attempts = $(L \times 0.25) = 0.25L$

Therefore, the total number of marks obtained = $(K - 0.25L)$

Now,

$$K + L = 75 \dots\dots\dots(i)$$

$$K - 0.25L = 56.25 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$1.25L = 18.75$$

$$L = 15$$

Now, put the value of L in equation (i), we get,

$$K + L = 75$$

$$K + 15 = 75$$

$$K = 60$$

Hence, the difference between the number of correct and wrong attempts made by the student in subject E = $(K - L) = (60 - 15) = 45$.

6. Questions

Subject A

The total number of questions attempted = 20P

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The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = $(17P \times 1) = 17P$

And, the total number of marks deducted from wrong attempts = $(3P \times 0.25) = 0.75P$

Therefore, the total number of marks obtained = $(17P - 0.75P) = 16.25P$

Now,

$$16.25P = (16P + 0.75)$$

$$0.25P = 0.75$$

$$P = 3$$

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = $(72 - 10.80Q)$

Now,

$$10.80Q - (72 - 10.80Q) = (7Q + 1)$$

$$3.80Q - 1 = 72 - 10.80Q$$

$$14.60Q = 73$$

$$Q = 5$$

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = $(72 - 10.80Q) = (72 - 54) = 18$

The total number of marks scored from correct attempts = $(54 \times 1) = 54$

And, the total number of marks deducted for wrong attempts = $(18 \times 0.25) = 4.50$

So, the total marks obtained = $(54 - 4.50) = 49.50$

Subject C

@QuizByMickey

The total number of questions attempted = $20R$

The number of correct attempts = $20R\%$ of $20R = 4R^2$

So, the number of wrong attempts = $(20R - 4R^2)$

Now,

$$4R^2 - (20R - 4R^2) = 48$$

$$8R^2 - 20R - 48 = 0$$

$$2R^2 - 5R - 12 = 0$$

$$2R^2 - 8R + 3R - 12 = 0$$

$$2R(R-4) + 3(R-4) = 0$$

$$(2R+3)(R-4) = 0$$

$$R = -1.50 \text{ or } 4$$

So, $R = 4$ (positive integer)

Therefore, the total number of questions attempted = $20R = 80$

The number of correct attempts = $20R\%$ of $20R = 4R^2 = 64$

And, the number of wrong attempts = $(20R - 4R^2) = (80 - 64) = 16$

Now,

The total number of marks scored from correct attempts = $(64 \times 1) = 64$

And, the total number of marks deducted for wrong attempts = $(16 \times 0.25) = 4$

So, the total marks obtained = $(64 - 4) = 60$

Subject D

The total number of questions attempted = $45S$

The number of correct attempts = 70% of $45S = 31.50S$

So, the number of wrong attempts = $(45S - 31.50S) = 13.50S$

Now,

The total number of marks scored from correct attempts = $(31.50S \times 1) = 31.50S$

And, the total number of marks deducted for wrong attempts = $(13.50S \times 0.25) = 3.375S$

So, the total marks obtained = $(31.50S - 3.375S) = 28.125S$

$$28.125S = 56.25$$

$$S = 2$$

@QuizByMickey

Therefore, the total number of questions attempted = $45S = 90$

The number of correct attempts = $31.50S = 63$

So, the number of wrong attempts = $13.50S = 27$

And, the total number of marks scored from correct attempts = $31.50S = 63$

And, the total number of marks deducted for wrong attempts = $3.375S = 6.75$

So, the total marks obtained = $(63 - 6.75) = 56.25$

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: B

Subject B (another student)

The number of correct attempts = $(54 - 4) = 50$

The number of wrong attempts = $(18 + 2) = 20$

So, the number of marks scored from correct attempts = $(50 \times 1) = 50$

And, the number of marks deducted from wrong attempts = $(20 \times 0.25) = 5$

Hence, the total number of marks obtained by another student in subject B = $(50 - 5) = 45$.

7. Questions

From the pie chart given above,

The total value of the pie chart = 100%

$$3K + K + (L + 10) + (L + 5) + L = 100$$

$$4K + 3L = 85 \dots\dots\dots(i)$$

Also,

$$3K / (L + 5) = 3/2$$

$$6K = 3L + 15$$

$$6K - 3L = 15 \dots\dots\dots(ii)$$

Solving, both the equations, we get,

@QuizByMickey

$$12K + 9L = 255$$

$$12K - 6L = 30$$

$$15L = 225$$

$$L = 15$$

Now, put the value of K in equation (i), we get,

$$4K + 3L = 85$$

$$4K + 45 = 85$$

$$4K = 40$$

$$K = 10$$

Now, the percentage distribution in the pie chart,

$$\text{Mid-Off} = 3K\% = 30\%$$

$$\text{Mid-On} = K\% = 10\%$$

$$\text{Mid-Wicket} = (L+10)\% = (15+10)\% = 25\%$$

$$\text{Square Leg} = (L+5)\% = (15+5)\% = 20\%$$

$$\text{Fine Leg} = L\% = 15\%$$

$$\text{The number of runs scored in mid-wicket} = (K+L+10) = (10+15+10) = 35$$

So,

$$25\% = 35$$

$$1\% = (35/25) = 1.40$$

$$100\% = (1.40 \times 100) = 140$$

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: B

According to the question,

The total number of runs scored by the player = 140

The total number of runs scored by the remaining 10 players of the team = $(26 \times 10) = 260$

So, the total number of runs scored by the whole team of 11 players = $(140 + 260) = 400$

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Therefore, the percentage of the number of runs scored by the player out of the total number of runs scored by the whole team of players = $\{(140/400) \times 100\} \% = 35\%$

So, $X = 35$

Hence, $2X = 70$.

8. Questions

From the pie chart given above,

The total value of the pie chart = 100%

$$3K + K + (L+10) + (L+5) + L = 100$$

$$4K + 3L = 85 \dots\dots\dots(i)$$

Also,

$$3K/(L+5) = 3/2$$

$$6K = 3L + 15$$

$$6K - 3L = 15 \dots\dots\dots(ii)$$

Solving, both the equations, we get,

$$12K + 9L = 255$$

$$12K - 6L = 30$$

$$15L = 225$$

$$L = 15$$

Now, put the value of K in equation (i), we get,

$$4K + 3L = 85$$

$$4K + 45 = 85$$

$$4K = 40$$

$$K = 10$$

Now, the percentage distribution in the pie chart,

$$\text{Mid-Off} = 3K\% = 30\%$$

$$\text{Mid-On} = K\% = 10\%$$

$$\text{Mid-Wicket} = (L+10)\% = (15+10)\% = 25\%$$

$$\text{Square Leg} = (L+5)\% = (15+5)\% = 20\%$$

$$\text{Fine Leg} = L\% = 15\%$$

$$\text{The number of runs scored in mid-wicket} = (K+L+10) = (10+15+10) = 35$$

So,

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$$25\% = 35$$

$$1\% = (35/25) = 1.40$$

$$100\% = (1.40 \times 100) = 140$$

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: C

The total number of runs scored = 140

The total number of balls faced = B

The overall strike rate = $(B+130)$

So,

$$(140/B) \times 100 = (B+130)$$

$$140/B = (B+130)/100$$

$$14000 = B^2 + 130B$$

$$B^2 + 130B - 14000 = 0$$

$$B^2 + 200B - 70B - 14000 = 0$$

$$B(B+200) - 70(B+200) = 0$$

$$(B-70)(B+200) = 0$$

$$B = 70 \text{ or } -200$$

So, B = 70 (positive integer)

Therefore, the overall strike rate = $(B+130) = (70+130) = 200$ @QuizByMickey

Hence, the required difference = $(200-140) = 60$.

9. Questions

From the pie chart given above,

The total value of the pie chart = 100%

$$3K + K + (L+10) + (L+5) + L = 100$$

$$4K + 3L = 85 \dots\dots\dots(i)$$

Also,

$$3K/(L+5) = 3/2$$

$$6K = 3L + 15$$

$$6K - 3L = 15 \dots\dots\dots(ii)$$

Solving, both the equations, we get,

$$12K + 9L = 255$$

$$12K - 6L = 30$$

$$15L = 225$$

$$L = 15$$

Now, put the value of K in equation (i), we get,

$$4K + 3L = 85$$

$$4K + 45 = 85$$

$$4K = 40$$

$$K = 10$$

Now, the percentage distribution in the pie chart,

$$\text{Mid-Off} = 3K\% = 30\%$$

$$\text{Mid-On} = K\% = 10\%$$

$$\text{Mid-Wicket} = (L+10)\% = (15+10)\% = 25\%$$

$$\text{Square Leg} = (L+5)\% = (15+5)\% = 20\%$$

$$\text{Fine Leg} = L\% = 15\%$$

$$\text{The number of runs scored in mid-wicket} = (K+L+10) = (10+15+10) = 35$$

So,

$$25\% = 35$$

$$1\% = (35/25) = 1.40$$

$$100\% = (1.40 \times 100) = 140$$

@QuizByMickey

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: A

Mid-Off

The total number of runs scored = 42

The number of runs scored through singles = $(1 \times 13) = 13$

The number of runs scored through doubles = $(2 \times 1) = 2$

The number of runs scored through triples = $(3 \times 1) = 3$

So, the total number of runs scored through singles, doubles and triples (excluding boundaries) = $(13+2+3) = 18$

Therefore, the total number of runs scored through boundaries (fours + sixes) = $(42-18) = 24$

Now, let the number of fours and sixes scored be F and S respectively,

So, the number of runs scored through fours = $(F \times 4) = 4F$

And, the number of runs scored through sixes = $(S \times 6) = 6S$

Now,

$$F + S = 5 \dots\dots\dots(i)$$

$$4F + 6S = 24 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$4F + 4S = 20$$

$$4F + 6S = 24$$

$$2S = 4$$

$$S = 2$$

Now, put the value of S in equation (i), we get,

$$F + S = 5$$

$$F + 2 = 5$$

$$F = 3$$

Hence, the total number of runs scored in mid-off through sixes only = $6S = 12$.

10. Questions

From the pie chart given above,

The total value of the pie chart = 100%

$$3K + K + (L+10) + (L+5) + L = 100$$

$$4K + 3L = 85 \dots\dots\dots(i)$$

Also,

$$3K/(L+5) = 3/2$$

$$6K = 3L + 15$$

$$6K - 3L = 15 \dots\dots\dots(ii)$$

Solving, both the equations, we get,

$$12K + 9L = 255$$

$$12K - 6L = 30$$

$$15L = 225$$

$$L = 15$$

Now, put the value of K in equation (i), we get,

$$4K + 3L = 85$$

$$4K + 45 = 85$$

$$4K = 40$$

$$K = 10$$

Now, the percentage distribution in the pie chart,

$$\text{Mid-Off} = 3K\% = 30\%$$

$$\text{Mid-On} = K\% = 10\%$$

$$\text{Mid-Wicket} = (L+10)\% = (15+10)\% = 25\%$$

$$\text{Square Leg} = (L+5)\% = (15+5)\% = 20\%$$

$$\text{Fine Leg} = L\% = 15\%$$

$$\text{The number of runs scored in mid-wicket} = (K+L+10) = (10+15+10) = 35$$

So,

$$25\% = 35$$

$$1\% = (35/25) = 1.40$$

$$100\% = (1.40 \times 100) = 140$$

Therefore, the total number of runs scored in all the fielding positions together = 140

$$\text{So, the number of runs scored in Mid-Off} = 30\% \text{ of } 140 = 42$$

$$\text{The number of runs scored in Mid-On} = 10\% \text{ of } 140 = 14$$

$$\text{The number of runs scored in Mid-Wicket} = 25\% \text{ of } 140 = 35$$

$$\text{The number of runs scored in Square Leg} = 20\% \text{ of } 140 = 28$$

$$\text{The number of runs scored in Fine Leg} = 15\% \text{ of } 140 = 21$$

Answer: D

@QuizByMickey

Fine Leg

The number of runs scored = 21

Strike rate = 210

Let, the number of balls faced be n .

So,

$$(21/n) \times 100 = 210$$

$$21/n = 210/100$$

$$210n = 2100$$

$$n = 10$$

Therefore, the total number of balls faced = 10

The number of dot balls faced = 3

So, the number of balls faced (excluding dot balls) = $(10-3) = 7$

Hence, the strike rate (excluding dot balls faced) = $\{(21/7) \times 100\} = 300$.

11. Questions

From the pie chart given above,

The total value of the pie chart = 100%

$$3K + K + (L+10) + (L+5) + L = 100$$

$$4K + 3L = 85 \dots\dots\dots(i)$$

Also,

$$3K/(L+5) = 3/2$$

$$6K = 3L + 15$$

$$6K - 3L = 15 \dots\dots\dots(ii)$$

Solving, both the equations, we get,

$$12K + 9L = 255$$

$$12K - 6L = 30$$

$$15L = 225$$

$$L = 15$$

Now, put the value of K in equation (i), we get,

$$4K + 3L = 85$$

$$4K + 45 = 85$$

$$4K = 40$$

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$$K = 10$$

Now, the percentage distribution in the pie chart,

$$\text{Mid-Off} = 3K\% = 30\%$$

$$\text{Mid-On} = K\% = 10\%$$

$$\text{Mid-Wicket} = (L+10)\% = (15+10)\% = 25\%$$

$$\text{Square Leg} = (L+5)\% = (15+5)\% = 20\%$$

$$\text{Fine Leg} = L\% = 15\%$$

$$\text{The number of runs scored in mid-wicket} = (K+L+10) = (10+15+10) = 35$$

So,

$$25\% = 35$$

$$1\% = (35/25) = 1.40$$

$$100\% = (1.40 \times 100) = 140$$

Therefore, the total number of runs scored in all the fielding positions together = 140

$$\text{So, the number of runs scored in Mid-Off} = 30\% \text{ of } 140 = 42$$

$$\text{The number of runs scored in Mid-On} = 10\% \text{ of } 140 = 14$$

$$\text{The number of runs scored in Mid-Wicket} = 25\% \text{ of } 140 = 35$$

$$\text{The number of runs scored in Square Leg} = 20\% \text{ of } 140 = 28$$

$$\text{The number of runs scored in Fine Leg} = 15\% \text{ of } 140 = 21$$

Answer: A

Another Player

$$\text{The total number of runs scored} = 4/7 \text{ of } 140 = 80 \text{ (Fractional value of } 42.85\% = 3/7)$$

Let, the total number of balls faced and the overall strike rate be 5m and 16m respectively.

So,

$$(80/5m) \times 100 = 16m$$

$$16/m = 4m/25$$

$$4m^2 = 400$$

$$m^2 = 100$$

$$m = 10$$

Hence, the total number of balls faced by another player = 5m = 50.

12. Questions

From the pie chart given above, we have,

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The total value of the pie chart = 360°

$$1.50a + 0.60b + (a+18) + (2b+6) = 360$$

$$2.50a + 2.60b = 336 \dots\dots\dots(i)$$

From the equation given above, we have,

$$P^2 - 3P - 108 = 0$$

$$P^2 - 12P + 9P - 108 = 0$$

$$P(P-12) + 9(P-12) = 0$$

$$(P+9) + (P-12) = 0$$

$$P = -9 \text{ or } 12$$

So, the difference between the values of a and b = 12 (positive root)

Now,

$$a - b = 12 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$2.50a + 2.60b = 336 - (i)$$

$$a - b = 12 - (ii)$$

$$2.50a + 2.60b = 336$$

$$2.50a - 2.50b = 30$$

$$5.10b = 306$$

$$b = 60$$

Now, put the value of b in equation (i), we get,

$$2.50a + 2.60b = 336$$

$$2.50a + 156 = 336$$

$$2.50a = 180$$

$$a = 72$$

So, the degree values in the pie chart for each college,

$$P = 1.50a^\circ = 108^\circ$$

$$Q = 0.60b^\circ = 36^\circ$$

$$R = (a+18)^\circ = (72+18)^\circ = 90^\circ$$

$$S = (2b+6)^\circ = (120+6)^\circ = 126^\circ$$

The sum of the total number of students studying in commerce stream in college P and college Q = (40×2)
= 80

@QuizByMickey

So,

$$(108+36)^\circ = 80$$

$$144^\circ = 80$$

$$360^\circ = \{(80/144) \times 360\} = 200$$

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

$$P = 108^\circ \text{ of } 200 = \{(108/360) \times 200\} = 60$$

$$Q = 36^\circ \text{ of } 200 = \{(36/360) \times 200\} = 20$$

$$R = 90^\circ \text{ of } 200 = \{(90/360) \times 200\} = 50$$

$$S = 126^\circ \text{ of } 200 = \{(126/360) \times 200\} = 70$$

Answer: C

According to the question,

College P

The total number of students studying in science stream = $(X-48)$

So, the total number of students studying in arts stream = $(X-48-62) = (X-110)$

The number of students studying in commerce stream = $X\% \text{ of } (X-110) = \{(X^2-110X)/100\}$

Now,

$$\{(X^2-110X)/100\} = 60$$

$$X^2 - 110X = 6000$$

$$X^2 - 110X - 6000 = 0$$

$$X^2 - 150X + 40X - 6000 = 0$$

$$X(X-150) + 40(X-150) = 0$$

$$(X+40) + (X-150) = 0$$

$$X = -40 \text{ or } 150$$

So, $X = 150$ (positive integer)

Hence, the total number of students studying in all the three streams together = $\{(X-48)+60+(X-110)\} = \{(150-48)+60+(150-110)\} = (102+60+40) = 202$.

13. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

@QuizByMickey

$$1.50a + 0.60b + (a+18) + (2b+6) = 360$$

$$2.50a + 2.60b = 336 \dots\dots\dots(i)$$

From the equation given above, we have,

$$P^2 - 3P - 108 = 0$$

$$P^2 - 12P + 9P - 108 = 0$$

$$P(P-12) + 9(P-12) = 0$$

$$(P+9) + (P-12) = 0$$

$$P = -9 \text{ or } 12$$

So, the difference between the values of a and b = 12 (positive root)

Now,

$$a - b = 12 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$2.50a + 2.60b = 336 - (i)$$

$$a - b = 12 - (ii)$$

$$2.50a + 2.60b = 336$$

$$2.50a - 2.50b = 30$$

$$5.10b = 306$$

$$b = 60$$

Now, put the value of b in equation (i), we get,

$$2.50a + 2.60b = 336$$

$$2.50a + 156 = 336$$

$$2.50a = 180$$

$$a = 72$$

So, the degree values in the pie chart for each college,

$$P = 1.50a^\circ = 108^\circ$$

$$Q = 0.60b^\circ = 36^\circ$$

$$R = (a+18)^\circ = (72+18)^\circ = 90^\circ$$

$$S = (2b+6)^\circ = (120+6)^\circ = 126^\circ$$

The sum of the total number of students studying in commerce stream in college P and college Q = (40×2)
 = 80

So,

@QuizByMickey

$$(108+36)^{\circ} = 80$$

$$144^{\circ} = 80$$

$$360^{\circ} = \{(80/144) \times 360\} = 200$$

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

$$P = 108^{\circ} \text{ of } 200 = \{(108/360) \times 200\} = 60$$

$$Q = 36^{\circ} \text{ of } 200 = \{(36/360) \times 200\} = 20$$

$$R = 90^{\circ} \text{ of } 200 = \{(90/360) \times 200\} = 50$$

$$S = 126^{\circ} \text{ of } 200 = \{(126/360) \times 200\} = 70$$

Answer: B

College S (Commerce Stream)

The total number of students studying = 70

The number of boys studying = 8Y

And, the number of girls studying = (8Y-Z)

Now,

$$(8Y-Z) = 3Z$$

$$8Y = 4Z$$

$$Y:Z = 1:2.$$

14. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

$$1.50a + 0.60b + (a+18) + (2b+6) = 360$$

$$2.50a + 2.60b = 336 \dots \dots \dots (i)$$

From the equation given above, we have,

$$P^2 - 3P - 108 = 0$$

$$P^2 - 12P + 9P - 108 = 0$$

$$P(P-12) + 9(P-12) = 0$$

$$(P+9) + (P-12) = 0$$

$$P = -9 \text{ or } 12$$

So, the difference between the values of a and b = 12 (positive root)

Now,

@QuizByMickey

$$a - b = 12 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$2.50a + 2.60b = 336 - (i)$$

$$a - b = 12 - (ii)$$

$$2.50a + 2.60b = 336$$

$$2.50a - 2.50b = 30$$

$$5.10b = 306$$

$$b = 60$$

Now, put the value of b in equation (i), we get,

$$2.50a + 2.60b = 336$$

$$2.50a + 156 = 336$$

$$2.50a = 180$$

$$a = 72$$

So, the degree values in the pie chart for each college,

$$P = 1.50a^\circ = 108^\circ$$

$$Q = 0.60b^\circ = 36^\circ$$

$$R = (a+18)^\circ = (72+18)^\circ = 90^\circ$$

$$S = (2b+6)^\circ = (120+6)^\circ = 126^\circ$$

The sum of the total number of students studying in commerce stream in college P and college Q = $(40 \times 2) = 80$

So,

$$(108+36)^\circ = 80$$

$$144^\circ = 80$$

$$360^\circ = \{(80/144) \times 360\} = 200$$

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

$$P = 108^\circ \text{ of } 200 = \{(108/360) \times 200\} = 60$$

$$Q = 36^\circ \text{ of } 200 = \{(36/360) \times 200\} = 20$$

$$R = 90^\circ \text{ of } 200 = \{(90/360) \times 200\} = 50$$

$$S = 126^\circ \text{ of } 200 = \{(126/360) \times 200\} = 70$$

Answer: D

College Q

The total number of students studying all the three stream together = 81

The number of commerce students studying = 20

Let, the difference between the number of commerce and arts students studying be v . (commerce-arts)

So, the number of arts students studying = $(20-v)$

Therefore, the number of science students studying = $(5v+1)$

Now,

$$20 + (20-v) + (5v+1) = 81$$

$$4v = 40$$

$$v = 10$$

Hence, the number of science students studying = $(5v+1) = (50+1) = 51$.

15. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

$$1.50a + 0.60b + (a+18) + (2b+6) = 360$$

$$2.50a + 2.60b = 336 \dots\dots\dots(i)$$

From the equation given above, we have,

$$P^2 - 3P - 108 = 0$$

$$P^2 - 12P + 9P - 108 = 0$$

$$P(P-12) + 9(P-12) = 0$$

$$(P+9) + (P-12) = 0$$

$$P = -9 \text{ or } 12$$

So, the difference between the values of a and $b = 12$ (positive root)

Now,

$$a - b = 12 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$2.50a + 2.60b = 336 - (i)$$

$$a - b = 12 - (ii)$$

$$2.50a + 2.60b = 336$$

$$2.50a - 2.50b = 30$$

$$5.10b = 306$$

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$$b = 60$$

Now, put the value of b in equation (i), we get,

$$2.50a + 2.60b = 336$$

$$2.50a + 156 = 336$$

$$2.50a = 180$$

$$a = 72$$

So, the degree values in the pie chart for each college,

$$P = 1.50a^\circ = 108^\circ$$

$$Q = 0.60b^\circ = 36^\circ$$

$$R = (a+18)^\circ = (72+18)^\circ = 90^\circ$$

$$S = (2b+6)^\circ = (120+6)^\circ = 126^\circ$$

The sum of the total number of students studying in commerce stream in college P and college Q = $(40 \times 2) = 80$

So,

$$(108+36)^\circ = 80$$

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$$P = 108^\circ \text{ of } 200 = \{(108/360) \times 200\} = 60$$

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$$R = 90^\circ \text{ of } 200 = \{(90/360) \times 200\} = 50$$

$$S = 126^\circ \text{ of } 200 = \{(126/360) \times 200\} = 70$$

Answer: A

Commerce Stream

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The number of students studying in college T = K

So, the total number of students studying in all the five colleges (including college T) together = $\{(K/1) \times 6\} = 6K$

Therefore, the total number of students studying in rest of the four colleges (excluding college T) together = $(6K - K) = 5K$

Now,

$$5K = 200$$

$K = 40$

Hence, the number of students studying in commerce stream in college $T = K = 40$.

16. Questions

Answer: A

The time taken to complete work by,

$$(A+B) = (40/7) \text{ days}$$

$$(C+D) = (60/11) \text{ days}$$

Total Work = 120
units (L.C.M. of time
taken)

Efficiency of,

$$(A+B) = \{120/(40/7)\} = \{(120 \times 7)/40\} = 21$$

$$(C+D) = \{120/(60/11)\} = \{(120 \times 11)/60\} = 22$$

Let, the efficiency of,

$$A = p$$

$$B = (21-p)$$

$$C = q$$

$$D = (22-q)$$

Now,

$$p/(22-q) = 60/100$$

$$p/(22-q) = 3/5$$

$$5p = 66 - 3q$$

$$5p + 3q = 66 \dots\dots\dots(i)$$

Also,

$$(21-p)/q = 5/4$$

$$84 - 4p = 5q$$

$$4p + 5q = 84 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$20p + 12q = 264$$

$$20p + 25q = 420$$

$$13q = 156$$

$$q = 12$$

Now, put the value of q in equation (i), we get,

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$$5p + 3q = 66$$

$$5p + 36 = 66$$

$$5p = 30$$

$$p = 6$$

So, the efficiency of,

$$A = p = 6$$

$$B = (21-p) = (21-6) = 15$$

$$C = q = 12$$

$$D = (22-q) = (22-12) = 10$$

Quantity I: The time taken by A and C to complete the work together = $\{(120)/(6+12)\} = (120/18) = (20/3) = 6.67$ days.

Quantity II: The time taken by B and D to complete the work together = $\{(120)/(15+10)\} = (120/25) = (24/5) = 4.80$ days.

Hence, Quantity I > Quantity II.

17. Questions

Answer: A

Speed of the boat,

In downstream = $(7X+2)$ km/h

In upstream = (X^2+2) km/h

Now,

$$\{240/(7X+2)\} / \{180/(X^2+2)\} = 4/5$$

$$\{1200/(7X+2)\} = \{720/(X^2+2)\}$$

$$1200X^2 + 2400 = 5040X + 1440$$

$$1200X^2 - 5040X + 960 = 0$$

$$5X^2 - 21X + 4 = 0$$

$$5X^2 - 20X - X + 4 = 0$$

$$5X(X-4) - 1(X-4) = 0$$

$$(5X-1)(X-4) = 0$$

$$X = 0.20 \text{ or } 4$$

So, $X = 4$ (positive integer)

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Therefore, the downstream speed = $(7X+2) = (28+2) = 30$ km/h

Upstream speed = $(X^2+2) = (4^2+2) = (16+2) = 18$ km/h

So, the speed of the boat in still water = $\{(30+18)/2\} = (48/2) = 24$ km/h

And, the speed of the river = $\{(30-18)/2\} = (12/2) = 6$ km/h

Quantity I: The speed of the boat in still water = 24 km/h

The speed of the river = 6 km/h

Hence, the required speed = $\{(24/6) \times 100\} \% = 400\%$.

So, Quantity I = 400

Quantity II: The downstream distance covered by the boat in 7 hours = $(7 \times 30) = 210$ km.

So, Quantity II = 210

Hence, Quantity I > Quantity II.

18. Questions

Answer: B

Shirt

CP = Rs. 500

SP = 120% of 500 = Rs. 600

Jeans

CP = $\{(600/3) \times 4\} =$ Rs. 800

MP = 140% of 800 = Rs. 1120

So, the MP of the shirt = $(1120-370) =$ Rs. 750

SP of the jeans = 90% of 1120 = Rs. 1008

Quantity I: MP of shirt = Rs. 750

Quantity II: SP of jeans = Rs. 1008

Hence, Quantity I < Quantity II.

19. Questions

Answer: B

The amount invested = Rs. 40000

The rate of interest = 5R% p.a. (S.I.)

The time duration = R years

So, the interest earned,

$(5R \times R)\%$ of 40000 = $5R^2\%$ of 40000 = Rs. $2000R^2$

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Now,

$$2000R^2 = 18000$$

$$R^2 = 9$$

$$R = 3$$

Quantity I

The amount invested = Rs. 30000

The rate of interest = $(R+7)\% = (3+7)\% = 10\%$ p.a. (C.I.)

The time duration = 2 years

$$\text{So, } [10+10+\{(10 \times 10)/100\}]\% = (20+1)\% = 21\%$$

The interested earned = 21% of 30000 = Rs. 6300.

Hence, Quantity I = 6300.

Quantity II

The amount invested = $8000R = \text{Rs. } 24000$

The rate of interest = 18% p.a. (S.I.)

The time duration = 4 years

So, the interest earned = 72% of 24000 = Rs. 17280.

Hence, Quantity I = 17280.

Therefore, Quantity I < Quantity II.

20. Questions

Answer: B

Let, the income of A and B be Rs. 7P and Rs. 6P respectively.

Also, the ratio of the expenditure of A to that of B = 100:90 = 10:9. Let, it be 10Q and 9Q respectively.

So, the savings of A = Rs. $(7P-10Q)$

And, the savings of B = Rs. $(6P-9Q)$

Now,

$$7P - 9Q = 10000 \dots\dots\dots(i)$$

$$6P - 10Q = 4000 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$42P - 54Q = 60000$$

$$42P - 70Q = 28000$$

$$16Q = 32000$$

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$$Q = 2000$$

Now, put the value of Q in equation I, we get,

$$7P - 9Q = 10000$$

$$7P - 18000 = 10000$$

$$7P = 28000$$

$$P = 4000$$

Quantity I: The expenditure of B = Rs. $9Q$ = Rs. 18000

And, the savings of Q = $(6P - 9Q)$ = $(24000 - 18000)$ = Rs. 6000

So, the required difference = $(18000 - 6000)$ = Rs. 12000

Hence, Quantity II = 12000.

Quantity II: The total income of A = $7P$ = Rs. 28000.

Hence, Quantity II = 28000.

Therefore, Quantity I < Quantity II.

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